

Assignment 3

BSSE23-A/B

Multivariable Calculus | (MT101T) | Fall '25

Name: _____ Reg. No.: _____

Total Points = 65

Q1. [CLO-3] [5+5pts]

Evaluate the double integral, where R is the region in the first quadrant enclosed by $xy = 1$, $y = x$, and $y = 2x$. Draw the region as well, by hand or with the help of a graphing utility.

$$\iint_R x^2 dA$$

Q2. [CLO-3] [5+5+5pts]

- a) By hand or with the help of a graphing utility, make a sketch of the region R enclosed between the curves
 $y = 4x^3 - x^4$ and $y = 3 - 4x + 4x^2$
- b) Find the intersections of the curves in part (a).
- c) Find

$$\iint_R x dA$$

Q3. [CLO-3] [5+5pts]

Use Cylindrical coordinates to find the volume of the solid that is bounded above by the sphere $x^2 + y^2 + z^2 = 1$ and below by the cone $z = \sqrt{x^2 + y^2}$. Draw the sketch as well, by hand or with the help of a graphing utility.

Q4. [CLO-3] [5+5pts]

Use a graphing utility to generate a plot of the vector fields.

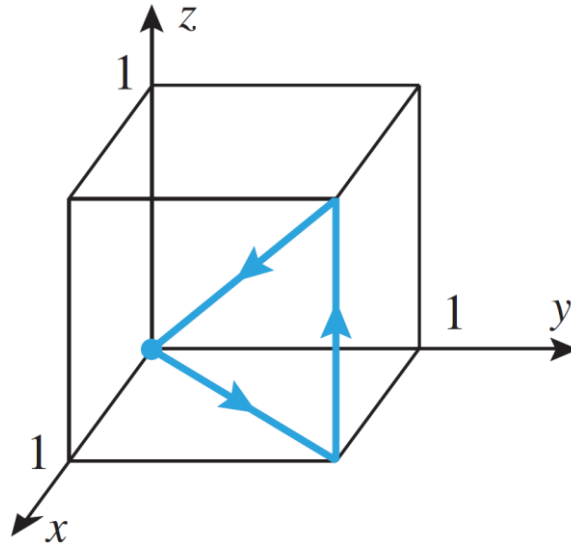
- a) $\mathbf{F}(x, y) = \mathbf{i} + \cos y \mathbf{j}$
- b) $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$

Q5. [CLO-3] [10pts]

Evaluate the line integral

$$\int_C x^2 z \, dx - yx^2 \, dy + 3dz$$

along the curve C shown in figure.



Q6. [CLO-3] [10pts]

Confirm that the force field $\mathbf{F}$ is conservative in some open connected region containing the points P and Q , and then find the work done by the force field on a particle moving along an arbitrary smooth curve in the region from P to Q .

$$\mathbf{F}(x, y) = xy^2 \mathbf{i} + x^2 y \mathbf{j}; P(1, 1), Q(0, 0)$$